

measurable metrics. My sCool Server was conceptualised and created to address this need. The interactive whiteboard is the latest improvement added to it, based on our ongoing assessments of user requirements.

Q. Is My sCool Server the company's first product or are there any other products as well?

A. My sCool Server is the first and only commercial product from the company. Though it was initially conceived to cater to the computer lab needs of the Gujarat State Board schools, today it has come a long way.

The Li-f-e: Linux for Education distribution on which the My sCool Server is built has been available as a free download since over a decade.

Q. Can you tell us more about the open source technologies you have used in building My sCool Server?

A. My sCool Server is built on off-the-shelf hardware – Intel NUC and open source technologies. Apart from the Li-f-e OS that is based on Ubuntu LTS, the Linux Terminal Server Project (LTSP) is integrated to serve 'fat clients' in lab computers. Also, the Epopotes teaching assistant is integrated so that teachers can screen broadcasts, as well as monitor and control each PC in the network. The Moodle LMS (learning management system), Kolibri to provide content from the Khan Academy, PHET, the CSpashshala curriculum, etc, and Sugarizer are integrated in the device. Spoken tutorial (spoken-tutorial.org) by IIT Mumbai, the National Repository of Open Educational Resources (NROER) initiative by the Central Institute of Educational Technology (CIET), and the NCERT are also made available offline for Indian schools.



Students at Tibetan Children's Village School in Suja, Himachal Pradesh, explore My sCool Server

“Creating a plug-and-play device that incorporates so many varied technologies on a Linux platform has been the biggest challenge, and overcoming this has been our greatest achievement”

Q. What were the biggest challenges that emerged initially while designing My sCool Server, and how did your team overcome them?

A. Our aim was clear about, “keeping things so simple that teachers and students are able to start using the system as soon as it is powered up, without having to go through manuals.” With My sCool Server, the basic operations are intuitive and advanced operations can be learnt without requiring extraordinary Linux systems administration know-how. We have designed a setup that configures

all services automatically depending on the environment the device is plugged into. Creating a plug-and-play device that incorporates so many varied technologies on a Linux platform has been the biggest challenge, and overcoming this has been our greatest achievement. We have also put together excellent and relevant tutorials and manuals to enable smooth operations for everyday use.

Li-f-e began as an openSUSE based distribution. After openSUSE discontinued the release of live media, we had to look for an alternative base. Ubuntu was an obvious choice, as most Indian state computer education curricula are designed for it. The shift from openSUSE to an Ubuntu base without compromising on the user experience was one of the big challenges.

We have also designed a snapshot backup system that can create snapshots on the device or create bootable USBs with all customisations at any point.

Making modules of offline content platforms was another challenge, since each content provider uses different technologies. Moodle, for example, uses the full LAMP stack. We created a self-contained Bitnami bundle to provide everything it needed so that any breakage due to system updates does not affect its services.